

Overview:

Computer Science student seeking opportunities to apply my programming skills and also learn from practical experiences in software development. Eager to learn from professionals within the industry and to be able to contribute to a team. I am looking to expand my knowledge in the fields of software development, data science, machine learning, artificial intelligence, design, and participating in research.

Education:

University of Texas Rio Grande Valley

Aug 2022-Current

Bachelor of Science in Computer Science –expected graduation:Dec 2025

GPA: 3.34/4.0

Skills:

Technical:

- C++ - Python - Linux/Unix - HTML - CSS - Javascript - Data Structure & Algorithms - Figma - React - Git - Django

Non-Technical:

- Bilingual: Fluent in Spanish and English

Relevant Courses :

Object Oriented Programming in Python

University of Texas Rio Grande Valley (Aug-Dec 2024)

Data Structures & Algorithms

(Aug-Dec 2024)

UI/UX Design

(Jan 2025 - Current)

Software Engineering I

(Jan 2025 - Current)

Projects:

Portfolio Website

-HTML, CSS, Javascript, Web Development

- Developed a responsive portfolio website using HTML, CSS, and Javascript with Visual Studio Code, deployed on GitHub Pages.

-Used and worked with Git/Github workflow for version management and collaborative development.

- Recent: Modified responsive design features to optimize display across various screen sizes and mobile devices.

Comparative Analysis of U-Net Variants for Brain Tumor 3D Segmentation

-Research, Python, Pytorch, Deep Learning

-Implemented UNETR and SegResNet neural networks for automated brain tumor segmentation using MONAI framework.

-Model performance evaluation on Medical Segmentation Decathlon dataset through quantitative metrics and visual analysis.

-Conducted comparative analysis of segmentation using IoU, Dice coefficient, and visual inspection.

-Learned about medical image processing, deep learning, and PyTorch implementations for healthcare research applications.

Amazon App Redesign

- Figma, UI/UX Design, App Development

-Amazon mobile shopping app redesign using Figma design tools.

-Enhanced user experience by reducing visual clutter and working on creating an intuitive shopping experience.

-Learned and developed experience in Figma for wireframing, prototyping, and collaborative design.

To-Do List Web & Desktop Applications

- Python, Django, HTML, CSS

-Developed a task management system with both a desktop and web interface using Python, Django, HTML, and CSS.

-Implemented a user authentication feature with login/registration functionality.

-Developed core features such as task creation, editing, deletion, date and deadline setting, and status tracking.

-Managed data consistency between desktop and web applications using Django.

Autocompleter

-C++, Algorithms and Data Structures

-Created an autocomplete system in C++ using AVL tree data structure.

-Implemented algorithm to find the top 3 most frequent string completions using a given set dictionary data.

-Worked on finding an optimized solution for search operations and applied tree concepts such as height-balancing and rotations.

Snake Game

-Python, Pygame, Object Oriented Programming

-Used Python to create a classic Snake Game using the Pygame library for real-time game mechanics.

-Implemented collision detection for boundary checking and self-collision to meet game requirements.

-Developed growing snake mechanics with food consumption and implemented score tracking.

-Applied responsive keyboard controls and added a game-over screen with the final score display.

Bank Account Management

- Python, OOP, Inheritance, Structures

-Developed a bank account management system in Python to practice OOP principles such as encapsulation and inheritance.

-Implemented a base bank account class with private attributes and public methods for deposits and withdrawals.

-Created subclasses for saving accounts and checking accounts with special functions like interests and withdrawal fees.

-Applied input validation and error handling to ensure data was properly managed.

Employment History:

H-E-B

Personal Shopper

Sep 2023-Current

Ensure timely order fulfillment by managing multiple orders simultaneously, meeting strict deadlines for curbside pickup.

Provide exceptional customer service by addressing customer inquiries, resolving issues promptly, and ensuring a positive shopping experience.

Organizations:

Society of Women Engineers

Member

Actively participated in SWE meetings, workshops, and events to promote the inclusion of women in engineering.